UNIVERSITY of GUYANA
FACULTY OF EDUCATION AND HUMANITIES
DEPARTMENT OF CURRICULUM AND INSTRUCTION

COURSE OUTLINE

COURSE NUMBER:	EMA 5104
COURSE NAME:	Methods of Teaching at the Secondary Level (Mathematics)
NUMBER OF CREDITS:	Three (3)
PRE & CO - REQUISITES:	EMA 5001
FOLLOW - ON COURSES:	Nil

COURSE DESCRIPTION:
Teaching mathematics at secondary institutions is changing. Fewer students are passing mathematics and opting for exclusively mathematical studies. Teachers must strive to raise their students' motivation, interest, determination, and desire to succeed if they are to encourage them to pursue and continue studying mathematics.

This course will enhance secondary-level mathematics teachers' pedagogical content knowledge. It will also serve as a theoretical base for practicum, which is conducted simultaneously.

STUDENT LEARNING OUTCOMES:
By the end of the course, students will:

- Develop individual's study skills through the preparation of materials for class presentation. (Micro- teaching)
- Develop insights, techniques and skills in the teaching and learning of mathematics and their development as competent and effective practitioners.

COURSE CONTENT:

Week 1: *Discourse on course outline*
(a) Reflection on positive and negative experiences as a student: Common mistakes teachers make; Things students respond to
(b) The nature of mathematics
(c) The aims of teaching mathematics

Week 2:
(a) Philosophy of teaching mathematics: What do you understand by the term *philosophy of teaching*? Outline your philosophy of teaching mathematics.
(b) Behavioural objectives
(c) Teaching model: Unit and lesson planning
Reflection (General)

Week 3:
(a) Effective and Efficient Schools
(b) Characteristics of an effective teacher
(c) Motivation and classroom management

Week 4:
(a) The nature of problem solving and problem-based learning
(b) Problems and benefits of a problem-solving approach
(c) Multiple Intelligences: Integrating into classroom practice
Presentation

Week 5:
(a) New emphasis in the teaching of Mathematics: creating a new teaching/learning environment.
(b) The constructivist perspective
(c) Inquiry Teaching
(d) Differentiation in the classroom.
Presentation

Week 6:
(a) Curriculum connections (links)
(b) Concept map as a cognitive tool
(c) Integrating Environmental Education into the Mathematics Curriculum
Presentation

Week 7: Mid-Semester Test

Week 8:
(a) Mathematical power: imagery, conjecturing, specialising and generalising.
(b) Questions and prompts for mathematical thinking

Week 9: Approaches to topics in secondary school mathematics:
Algebra(1) and Function/Pre-Calculus

Presentation

Week 10: Approaches to teaching topics in mathematics:
Geometry and Trigonometry
Reflection

Week 11: Approaches to teaching topics in mathematics:
Statistics and Probability
Presentation

Week 12:
(a) Parent involvement in education
(b) Assessment procedure in mathematics classrooms

Week 13:
(a) Introduction to STEM Schools
(b) Technology and resources for teaching and learning mathematics
(c) Revision

Week 14: Final Test

Method of Teaching

Lecture, discussion, and analysis of materials introduced by the lecturer, group and individual presentations

Method of Assessment

Semester paper (1)	25% (Individual-5% for each week after the due date)
Presentations (2)	25% [Individual (10%+15%)]
Test (2)	40% (20% + 20%)
Class participation	10% (Reflection 5% + Class discussion 5%)

REQUIRED READINGS:

1. Anderson, K. & Minke, K. (2007). Parent Involvement in Education: Toward an understanding of parent's decision making, *Journal of Educational Research*, *100* (5),311-323.
2. Cangelosi, J. (2003).*Teaching Mathematics in Secondary and Middle School. An interactive approach.* (3rd ed.). Merrill Practice Hall
3. Joong, P., Baker-Gibson, J., Chai, A., Ramsawak-Jodha, N., Welsh, D., &Wintz, P. (2016). STEM Strategies in Secondary Schools. In B.J. Ogunkola & S. Blackman, (Ed.), *Transforming Learning: International Perspectives* (pp.157-197). Cambridge Scholars Publishing.
4. Mason, J. & Watson, A. (1998). *Questions and Prompts for Mathematical Thinking* Shaftesbury,Derby: Association of Teachers of Mathematics (ATM),
5. Stone, R. (2010). *Best Practices for High School Classrooms. What Award-Winning Secondary Teachers Do.* Corwin Press Inc
6. Wintz, P. (2008). An investigation into the extent to which mathematics student-teachers transfer their knowledge of constructivism to their classroom practice. *The Journal of Education and Humanities. 1*(1), pp.71-89.
7. Wintz,P., Joong, P. & Wintz, G. (2020). Integrating of environmental education into the mathematics curriculum: Effects on pupils' performance and environmental awareness. *Journal of the Faculty of Education and Humanities*, *1*, 92-109.

RECOMMENDED READINGS:

8. Journal for Research in Mathematics Education (2020 – present). National Council of Teachers of Mathematics.
9. Mathematics Teacher Journal (2020 – present). National Council of Teachers of Mathematics
10. The Journal of Mathematics Education. (2020 – present). National Council of Teachers of Mathematics
11. School Science and Mathematics. (2020 – present). School Science and Mathematics Association